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Adoption of Digital Technologies by SMEs for Sustainability and Value Creation: Moderating Role of Entrepreneurial Orientation

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Abstract: Digital business transformation is considered an effective business strategy that appears to have gained attention since the enterprises are challenged to continuously improve their business practices as well as capabilities. The use of digital technologies could reduce the influence of external crises and could introduce massive changes in business operations by providing better business models. Moreover, adopting digital technology can influence both economic sustainability and social value of enterprises and can improve regional socio-economic conditions. There are few recent studies on how technology can empower enterprises at different phases of growth and sustainability; furthermore, very few studies are available that determine how adopting different modern digital technologies can create value for small and medium enterprises (SMEs). Therefore, this study aims to close this gap and investigate the moderating role of entrepreneurial orientation. With the support of resource-based view (RBV) and dynamic capability view (DCV) theories, along with a literature review, a theoretical model has been developed. It was then validated using the PLS-SEM technique considering 319 respondents who are SME employees in India. The findings show that adopting digital technologies has a significant impact on the creation of economic sustainability and social value for SMEs. The study also found a significant moderating impact of entrepreneurial orientation on the relationship between social and economic value creation and SME performance.

Keywords: digital technology; entrepreneurship; economic and social sustainability; SME performance; entrepreneurial orientation

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1. Introduction

The accelerated growth of digital technologies and their adoption by enterprises have drastically changed the overall landscape of entrepreneurial activities and have impacted regional development. The present research on entrepreneurship, innovation, and the emergence of modern technologies, has thrown light on how technology-empowered enterprises are trying to reshape their entrepreneurship activities and drive their growth [1]. There are studies on how modern digital technology can help enterprises of different sizes, especially small- and mid-size enterprises (SMEs), can sustain their growth momentum, accelerate their business activities, and contribute to regional development [2–4]. Bartik, Bertrand, and Cullen [5] showed how SMEs could suffer from various constraints, but they are able to survive and adjust to external shocks. Welter, Baker, and Wirsching [6] observed that the abilities of different-sized enterprises to innovate and grow are closely related to their specific nature. Such innovation and growth are found to depend on institutional and regional contexts [7].

However, scholars have not been clear about the contributions of digital technologies such as social media, the Internet of Things (IoT), blockchain, big data analytics, and artificial intelligence (AI) enabled applications towards the growth of start-up SMEs and

related entrepreneurial activities. Scholars and practitioners are also uncertain how, when, and under what specific conditions the use of digital technologies could help start-up enterprises sustain their growth continuum [8]. This dilemma among the experts has been multiplied by the emergence of the COVID-19 pandemic when the start-up enterprises were unsure about how to restructure their business resilience using digital technology to trace out ways in which they could successfully retain potential customers in such an apocalyptic situation. This predicament also affects SMEs who need to change their business operations with modern digital technologies to create resilient economic and social values [9].

Reaping the benefits from using digital technologies does not support a "one-size-fits-all" approach, which engages scholars in more in-depth research on start-up enterprises and changes their notion of entrepreneurial activities [10]. In this context, it is argued that start-up SMEs suffering from various constraints need to use their existing resources and opportunities in the best possible way to extract the best outcomes by appropriately using modern digital means. This concept corroborates the resource-based view (RBV) theory [11].

It is perceived that start-up SMEs also could embolden their abilities to use digital technologies appropriately to sense and seize the available opportunities and then reconfigure their available resources to successfully address the volatile high-velocity business environments. This concept is in consonance with the dynamic capability view (DCV) theory [12].

The present study posits that when start-up SMEs appropriately use modern digital technologies, through effective adoption, they can improve the economic and social values of their region. By creating social and economic value, start-up SMEs can improve their performance provided they are equipped with appropriate entrepreneurial orientation. Studies on how the underserved start-up small business enterprises could benefit by using digital technologies are found to be limited and at the rudimentary stage [13]. There are not many studies that focus on the influence of modern digital technologies on creating economic and social value for a particular region. Thus, there is a research gap. Against such a background, the aim of this study is to address the following research questions.

- RQ1: How can the adoption of different digital technologies by SMEs influence value creation (both social and economic value)?
- RQ2: Can creating both social and economic value influence SME performance?
- RQ3: Does entrepreneurial orientation play any moderating role in influencing the relationship between value creation and SME performance?

2. Literature Review

Junaidah [14] found that SMEs are effective contributors to employment generation, export activities, and regional development. Start-up SMEs are valued for their impact on social and economic aspects, as well as for developing a nation's economic health by enriching inventions as well as innovations [15,16]. In developing countries, SMEs are seen as a dynamic, vibrant sector for economic growth [17,18]. Studies have demonstrated that SMEs may be considered effective drivers in alleviating poverty for a particular region [4,19].

The use of different digital technologies, such as social media platforms, blockchain, AI-enabled business applications, the Internet of Things, and big data analytics, can reduce the exogenous crisis faced by start-up entrepreneurs [20,21]. Digital technology adoption by start-up SMEs can influence both economic and social value, and it can improve the socio-economic conditions of that region [7,22]. The use of digital technologies can also impact the performance of enterprises, but the growth rate of enterprises that use technological applications has been found to vary [23,24].

Junaidh [14] and Ng et al. [17] have emphasized the detailed study to conceptualize how the enterprises looking for digital transformation could best integrate these ground-breaking technologies and could reestablish their models of operations through the usage of new, more advanced technologies. In this way, SMEs can do business by enhancing their

preparedness to overcome any crisis and by rethinking the implications of using digital technologies for their success in the emerging digital economy. Laverne et al. [25] have also suggested that a comprehensive and academically rigorous study is needed to understand how the use of digital technologies could facilitate interactions among the economic actors and create economic and social value. The study has also highlighted that IoT devices have possessed security and privacy issues in the context of authentication, heterogeneity, and identification, whereas big data usage also possesses some challenges, including poor data quality and data silos as well as a dearth of skills of the users, and these challenges could be addressed by the appropriate recruitment of skilled employees and by imparting proper training to the existing employees to improve their skillsets [26].

The adoption of modern digital technologies can trigger the digital transformation of start-up SMEs. Digital transformation refers to helping enterprises radically improve their performance by improving economic and social value [27]. Digital transformation is a way to change business activities from traditional means to a digital environment. For example, digitally transforming the retail industry could take the concept of "bricks and mortar" and make it a "clicks and bricks" environment [28].

What to speak of the SMEs, even the start-up SMEs are also trying to digitally transform their business practices by introducing massive changes in their business operations, ensuring better customer services, superior business models, payments with the new methods with online engagements through proper utilization of AI-enabled business applications, big data analytics, IoT, social media, blockchain, and other technologies [29]. Such integration of digital technologies is perceived to improve economic and social value, which impacts the overall performance of SMEs through an advanced way of doing business [30,31]. This concept received support from Sebestian et al. [32], who documented that social media, AI, and other digital technologies are fundamental driving forces for the digital transformation of enterprises to improve economic and social value, improve performance, and accelerate regional development.

3. Theoretical Underpinning and Hypothesis Development

3.1. Theoretical Underpinning

In order to elucidate how SMEs that adopt digital technology improve their performance by enhancing their economic and social value, this study has taken help from the resource-based view (RBV) theory [11] and dynamic capability view (DCV) theory [12]. To achieve a better competitive advantage, SMEs need to implement a strategy that cannot be easily replicated by their competitors [11]. How start-up SMEs can leverage resources to create and sustain competitive advantage by improving their performance has become the focus of research scholars as well as practitioners. Moreover, SMEs perform differently from one another as they have distinct capabilities and resources which are valuable, rare, inimitable, as well as non-substitutable (VRIN). This concept corroborates with RBV theory [11].

Using applications of social media, as well as other digital technologies, is considered part of the resource portfolio of start-up enterprises, but it can hardly meet the criteria of resource-based theory on its own due to the comparatively low barriers for other SMEs to acquire such applications. Thus, applying different digital technologies cannot really enhance the value of SMEs, and so it cannot strictly act as a VRIN resource [33]. Thus, in terms of the RBV theory, when SMEs simultaneously use other technologies such as AI, big data analytics, IoT, and blockchain, their performance will be superior to their counterparts in the identical market.

SMEs need to understand that the market environment and customer demands are changing rapidly. In such a dynamic market environment, SMEs need to sense, seize, and transform the available opportunities and external resources appropriately so that they can effectively react and respond to the changing demands of the customers. This could help the SMEs to outperform their competitors. In this context, the different digital technologies are deemed to enhance the abilities of the SMEs to effectively integrate, build, as well as reconfigure internal and external abilities for successfully addressing any challenges

which may emerge due to the rapidly changing business environment. This is the principal concept of dynamic capability view (DCV) theory [12].

Start-up SMEs can effectively improve their dynamic capability by effectively using digital technologies to orchestrate and reconfigure their competencies in the dynamic market environment. It is important that the SMEs have a proper strategy to compete with their competitors working in the same market [34]. In this context, entrepreneurial orientation seems to play a critical role in transforming the operational activities of SMEs and in improving their performance [35].

3.2. Social Media Application (SMA)

Paris, Lee, and Seery [36] (p. 531) defined social media marketing technology as "a secured generation of web-development and design, that aims to facilitate communication, sources, information sharing, interoperability, and collaboration on the World Wide Web". It has been ascertained that social media applications are very popular among younger people who spend a considerable amount of time on them [37]. Social media platforms are tools to easily create online communication between customers and SMEs [38,39]. Walsh and Lipinski [40] observed that SMEs used social media platforms to improve their brand-building activities, and Ware [41] also found that SMEs use them to help develop their business activities. In a study by Abed, Dwivedi, and Williams [42], it was observed that, in Saudi Arabia, SMEs used social media platforms for their electronic commerce. Thus, the use of social media platforms by SMEs is perceived to impact their economic value and social value. Accordingly, the following hypotheses are formulated.

H1a. *Social media application (SMA) positively impacts on the creation of economic value (ECV) for SMEs.*

H1b. *Social media application (SMA) positively impacts on the creation of social value (SOV) for SMEs.*

3.3. AI-Enabled Applications (AEA)

Studies have found that SMEs apply AI technologies to help them remodel various business activities such as supply chain networks, production systems, and operational management systems [43,44]. SMEs can remodel business applications with the help of AI technology according to their needs without incurring much cost [45]. AI technology involves machines that can efficiently perform like human beings [46].

Large enterprises, as well as SMEs, have used AI to develop their business activities. For example, KPMG is using AI technology to automate their auditing services, while Bridgewater Associates uses it to improve their business operational activities [47]. SMEs that specialize in finance, marketing, and telecommunication have long been using AI technology to enhance their competitive advantage [48].

AI-enabled applications are considered VRIN resources that impact production systems to make SMEs more competitive. This concept is supplemented by RBV theory [11]. AI is a technology that possesses a human-like intellect to perform complex tasks [49]. AI-enabled applications are thus perceived to influence the creation of an enterprise's economic as well as social value. In terms of the above discussions, the following hypotheses are formulated.

H2a. *AI-enabled applications (AEA) positively impact on the creation of economic value (ECV) for SMEs.*

H2b. *AI-enabled applications (AEA) positively impact on the creation of social value (SOV) for SMEs.*

3.4. Big Data Analytics (BDA)

Data analytics has gained huge momentum in recent years, consequently from the emergence of big data. Big data analytics (BDA) is a "holistic process that involves the

collection, analysis, use, and interpretation of data for various functional divisions with a view to gaining actionable insights creating business value and establishing the competitive advantage" [50] (p. 178). Traditional methods of performing analytics differ from BDA on four salient dimensions, which are variety, velocity, volume, and accessibility [51]. Owing to the dynamic characteristics of big data, velocity is referred to as the rate at which data are generated and analyzed, and it sometimes includes real-time analysis. Accessibility is construed as the ability of SMEs to collect data from multifarious sources [52]. It is important to mention here that when data remain in an unprocessed form, they have no value until they are examined with an appropriate analytical tool for extracting meaningful information. Thus, the application of BDA is perceived to impact SMEs towards the creation of economic as well as social value. Accordingly, the following hypotheses are developed.

H3a. *Big data analytics (BDA) positively impacts on the creation of economic value (ECV) for SMEs.*

H3b. *Big data analytics (BDA) positively impacts on the creation of social value (SOV) for SMEs.*

3.5. IoT Applications (IOA)

Recently, big data have experienced further growth with the emergence of the Internet of Things (IoT) technology. IoT includes machine intelligence, network technologies, as well as smart devices which are interconnected. In such a context, IoT can facilitate the constant, rapid exchange of data in a real-time scenario [53] and improve the upscaling process. This functionality leads to the generation of new and better products as well as services [54–56]. IoT is construed as an internet-embedded device [1]. The adoption of IoT technology is perceived to be essential for tracking indoor assets as well as outdoor assets [57]. SMEs can benefit from IoT applications to optimize their floor operations, improve sustainability in production, and update product-logistic operations [33]. Applications of IoT also help SMEs to sense, seize, and reconfigure external opportunities for their benefit [58,59]. This concept corroborates the DCV theory [12]. Devices that use IoT technology are associated with the EPC (electronic product code) network, which can provide a scalable information system that helps SMEs to exchange information exchange [60]. Such dynamic ability of IoT applications is perceived to help SMEs to create economic as well as social value. Accordingly, it is hypothesized as follows.

H4a. *IoT applications (IOA) positively impact on the creation of economic value (ECV) for SMEs.*

H4b. *IoT applications (IOA) positively impact on the creation of social value (SOV) for SMEs.*

3.6. Blockchain Applications (BCA)

Blockchain is considered a digital ledger that presents the detailed history of various transactions that are distributed over several computers, which are called "nodes" and which are duly operated by different participants [61]. This process allows the participants to introduce records that are supported by validated as well as immutable cryptographic protection [62]. Blockchain is considered to function like a distributed open service database [63] that uses advanced cryptography. BCA can never be hacked, and, from that perspective, it is considered a trusted platform [64]. BCA decentralizes user data, and it is gaining consensus as public networks of several participants use it to ensure information accuracy [65]. BCA can be used by SMEs to enhance their information security as well as to protect the data of their customers [66,67]. Thus, SMEs that apply blockchain technology are expected to see an impact on the creation of economic and social value. In such a scenario, the following hypotheses are developed.

H5a. *Blockchain applications (BCA) positively impact on the creation of economic value (ECV) for SMEs.*

H5b. *Blockchain applications (BCA) positively impact on the creation of social value (SOV) for SMEs.*

3.7. Economic Value (ECV)

Economic value (ECV) is considered the value that an enterprise always wants to derive from its available resources. One of the main objectives of SMEs is the creation of economic value through profit maximization [31]. SMEs have several ECV implications for SMEs to improve their bottom line [68]. ECV is considered a measure of benefits that are produced by a good or service for the economic agent [69]. The ECV is normally estimated with currency units. Purchasing a product brings economic value when it benefits the seller [70]. Businesses can create economic value by making profits. ECV is the maximum amount that someone is found willing to pay towards purchasing a good or service.

Different modern business applications could help to achieve better economic value for SMEs. The economic value changes if the price of the good or the service changes [71]. Modern business applications can help to minimize product costs and thereby can improve the profitability of SMEs [72]. If the price of a product increases very much, the potential customer may not purchase the product. In that case, the ECV decreases [73]. Thus, ECV is perceived to impact SME performance. Accordingly, it is hypothesized as follows.

H6. *Economic value (ECV) positively impacts SME performance (SMP).*

3.8. Social Value (SOV)

The creation of social values (SOV) by SMEs is considered to be the extent to which they have performed their work to benefit society. Social value emerges from the concept of corporate social responsibility (CSR) [74]. Customers may favor those SMEs that spend more to uplift society [75] by preferring to buy their products or services (Santos, 2011). Different digital applications can help to improve the social value of SMEs. The concept of social value is perceived to impact the overall performance of SMEs. Social values are normally developed by the SME leadership, and then they are accepted and adopted by the employees [37,74]. Social values are shared values among the employees who perceive those social values are considerably important, and enterprises also call them core values. Thus, social value is perceived to impact the performance of SMEs. Accordingly, the following hypothesis is derived.

H7. *Social value (SOV) positively impacts SME performance (SMP).*

3.9. Moderating Role of Entrepreneurial Orientation (EO)

Entrepreneurial orientation (EO) is considered an overall strategic posture of the enterprise [35]. In the context of enterprise growth, EO is expected to guide SME entrepreneurs to deploy modern applications early enough to improve business operations as the business environment changes. The EO is considered to be proactive and helps entrepreneurs to take the necessary steps toward creating innovative products and services. With the help of EO, SMEs will be able to enjoy advantages related to availing themselves of high-risk opportunities [76,77]. Studies demonstrate that EO has a positive impact on business growth in developing as well as developed countries [78]. Other studies also have highlighted that the relationship between EO and the growth of SMEs is positive [79]. EO could influence the relationship between value creation as well as enterprise performance and is thus perceived to influence SME performance. Accordingly, it is hypothesized as follows.

H8a. *Entrepreneurial orientation (EO) moderates the relationship between economic value (ECV) and SME performance (SMP).*

H8b. *Entrepreneurial orientation (EO) moderates the relationship between social value (SOV) and SME performance (SMP).*

With all these inputs, a conceptual model is developed, which is shown in Figure 1.

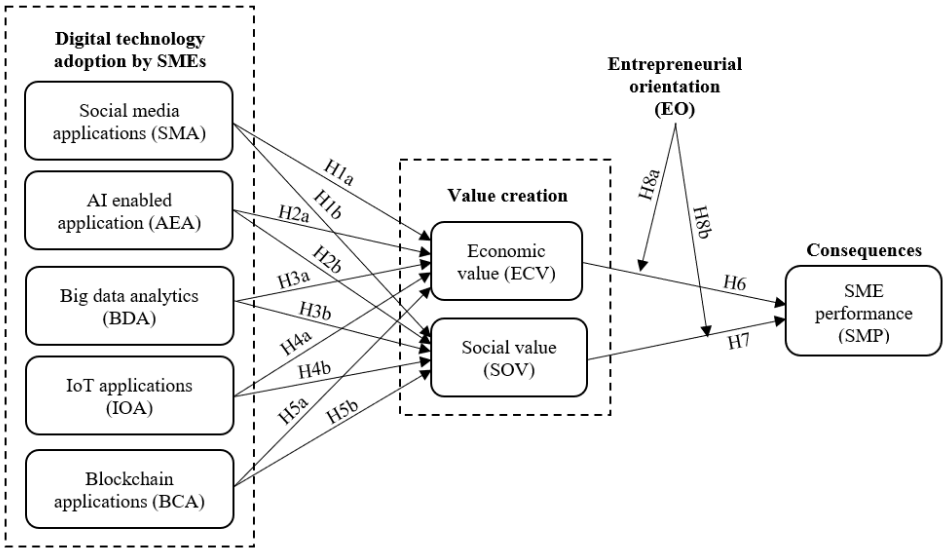


Figure 1. The conceptual model (Adopted from RBV and DCV theories).

4. Research Methodology

In order to test the hypotheses and validate the conceptual model, the data were analyzed with the partial least square structural equation modeling technique because this approach is simple and can analyze an exploratory study such as this [80]. With this method, it is easy to analyze the data, which are not normally distributed [81], which covariance-based structural equation modeling technique cannot do [82]. This process involved conducting a survey to obtain feedback from respondents. The responses were then quantified on a standard 5-point Likert scale with anchors at Strongly Disagree (SD) as 1 and Strongly Agree (SA) as 5.

4.1. Research Instruments

To prepare the survey questions that would be provided to the respondents, the authors used the help of extant literature and adjusted the questions to be appropriate to the context of the present study. The questions were prepared in the form of statements. Then, a pretest was conducted with a convenience sample of 30 respondents. From the outcomes of the pretest, the statements were rectified to enhance their understandability. After the pretest stage, a pilot test was conducted to ascertain content validity of the items and to enhance the readability of the questions. The pilot test analyzed the feedback from respondents who were knowledgeable about the area of this study and who did not participate in the main survey. With the results of the pilot test, the authors were able to ascertain the content validity of the items and modify the recitals of the questions to enhance their readability so that the prospective respondents may not have any difficulty in replying. After the pilot test, some experts with adequate knowledge in the domain of the present study were consulted for their opinions to enhance the comprehensiveness of the questionnaire statements. By following these steps, the authors were able to finetune 33 questions. Details of 33 questions are provided in Appendix A with their sources. Be it mentioned here, the questionnaire provided in the appendix is applicable to the different types of SMEs, including the start-up SMEs, and can be responded to by the different hierarchy of the managers such as senior managers, midlevel managers, junior managers, as well as the non-managerial employees of the SMEs.

4.2. Data Collection Strategy

The present study aimed to investigate the contributions of digital technologies toward SME performance. Thus, in the survey, data needed to be collected from respondents who possessed at least a basic concept of digital technologies and their contributions to SMEs. In such context, purposive sampling was deemed to be the correct method [83]. With

this process, the researchers depend principally on their own judgment for targeting the potential respondents. Since most of the authors of this study are based out of India, they deemed to be convenient to target the respondents from India. Hence, they preferred convenient sampling [84] along with purposive sampling.

For collection of inputs from the respondents, the authors attended some seminars as well as conferences held in different cities in India during the period from January 2022 to March 2022. The topics of discussion in these seminars and conferences covered the pros and cons of the adoption of digital technologies by SMEs in India for their sustainability and value creation. At those seminars and conferences, it was possible to contact some resource people who helped to supply details of prospective respondents who might agree to participate in the survey. The total number of such potential respondents was 807.

Those potential respondents were provided with the response sheets containing the 33 questions in the form of statements. Each respondent would answer the questions by putting one tick mark in one of the five options. Along with the response sheet, a guideline on how to fill in the response sheet was also provided. The respondents were also assured that their confidentiality and anonymity would be strictly preserved. The respondents were requested to reply within two months, and within the stipulated time, 331 responses were received. The response rate was 41.01%. On scrutiny of these 331 responses, it was found that 12 responses were incomplete and were, therefore, not considered. The statistical analysis was performed with the inputs of 319 respondents against 33 items, which is within the allowable range [85]. These 319 respondents consist of male and female managers holding different ranks in these SMEs, which have either adopted digital technologies or have been contemplating adopting digital technologies, and these SMEs are based out of India. The respondents, as such, are deemed to be conversant about how digital technology adoption could help the SMEs for sustainability and for value creation. Demographic statistics of these 319 respondents are provided in Table 1.

Table 1. Demographic statistics (n = 319).

Particulars	Category	Frequency	Percentage (%)
Gender	Male	252	78.9
	Female	67	21.1
Hierarchy	Senior manager	29	9.0
	Midlevel manager	83	26.0
	Junior manager	166	52.0
	Non-managerial employees	41	13.0

5. Analysis of Data and Results

In order to ascertain the content validity of the items, the loading factor (LF) of each item was computed. Then, to examine the validity, reliability, and the internal consistency of average variance extracted (AVE), composite reliability (CR), and Cronbach's alpha (α) has duly been estimated. The values of these constructs were found to be within the permissible range. The results are provided in Table 2.

It has been observed that the square roots of all the AVEs are greater than the corresponding bifactor correlation coefficients, which satisfies the Fornell and Larcker criteria [86]. This confirms the discriminant validity of the constructs. The results are provided in Table 3.

Table 2. Measurement properties.

Constructs/Items	LF	AVE	CR	α	*t*-Values
SMA		0.78	0.82	0.86	
SMA1	0.87				27.11
SMA2	0.85				22.16
SMA3	0.94				26.12
SMA4	0.87				33.01
AEA		0.88	0.92	0.95	
AEA1	0.90				32.11
AEA2	0.92				21.26
AEA3	0.97				25.18
AEA4	0.96				28.17
BDA		0.80	0.84	0.88	
BDA1	0.93				41.11
BDA2	0.89				27.29
BDA3	0.85				23.91
BDA4	0.96				22.06
IOA		0.92	0.95	0.98	
IOA1	0.90				26.11
IOA2	0.96				27.28
IOA3	0.87				32.17
IOA4	0.92				26.19
BCA		0.83	0.87	0.91	
BCA1	0.96				27.05
BCA2	0.89				35.31
BCA3	0.93				39.11
BCA4	0.85				33.46
ECV		0.75	0.78	0.82	
ECV1	0.90				35.17
ECV2	0.78				30.11
ECV3	0.85				32.48
ECV4	0.87				37.07
ECV5	0.91				36.14
SOV		0.85	0.88	0.92	
SOV1	0.85				26.12
SOV2	0.95				23.29
SOV3	0.96				39.83
SOV4	0.90				27.91
SOV5	0.94				32.78
SMP		0.81	0.84	0.89	
SMP1	0.80				21.12
SMP2	0.95				26.29
SMP3	0.85				30.57

Table 3. Discriminant validity test (Fornell and Larcker criteria).

Constructs	SMA	AEA	BDA	IOA	BCA	ECV	SOV	SMP	AVE
SMA	0.88								0.78
AEA	0.26	0.94							0.88
BDA	0.19	0.23	0.89						0.80
IOA	0.31	0.32	0.19	0.96					0.92
BCA	0.17	0.36	0.28	0.32	0.91				0.83
ECV	0.38	0.33	0.23	0.39	0.18	0.87			0.75
SOV	0.32	0.19	0.36	0.37	0.37	0.24	0.92		0.85
SMP	0.34	0.17	0.40	0.29	0.29	0.38	0.22	0.90	0.81

5.1. Moderator Analysis (Multigroup Analysis, MGA)

In the present study, the moderating effects of entrepreneur orientation (OE) were examined on the two linkages ECV→SMP (H6) and SOV→SMP (H7). While examining the

effects of EO on H6 and H7, the effects were categorized into two groups: Strong EO and Weak EO. In order to examine these effects, multigroup analysis (MGA) was performed with the bootstrapping procedure considering 5000 resamples. It is known that if the *p*-value difference for the effects of two categories of a moderator on a specific linkage is either greater than 0.95 or less than 0.05, then the effects of that moderator on that specific linkage are considered significant [87]. The results are shown in Table 4.

Table 4. Moderator analysis (MGA).

Linkage	Moderator	Hypothesis	*p*-Value Difference	Remarks
(ECV→SMP) × EO	EO	H8a	0.03	Significant
(SOV→SMP) × EO	EO	H8b	0.01	Significant

5.2. Hypotheses Testing

With consideration of separation distance 7, cross-validated redundancy for each dependent construct was assessed by estimating the Q^2 value, which came out to be 0.072 (positive) [88]. The result indicates that the model has predictive relevance.

In order to find out if the model is fit, the standardized root mean square residual (SRMR) was considered as a standard index for model validation. The values of SRMR emerged as 0.066 and 0.031 for PLS and for PLSc, respectively. These two values are greater than 0.08 [89]. This confirms that the model is in order. This procedure helped to estimate the path coefficients of different linkages, *p*-values, as well as R^2 values. The results are shown in Table 5.

With these inputs, the validated model is shown in Figure 2.

Table 5. Structural equation modeling.

Linkages	Hypotheses	Path Coefficients	*p*-Values	Remarks
SMA→ECV	H1a	0.17	$p < 0.01$ (**)	Supported
SMA→SOV	H1b	0.37	$p < 0.001$ (***)	Supported
AEA→ECV	H2a	0.19	$p < 0.05$ (*)	Supported
AEA→SOV	H2b	0.36	$p < 0.01$ (**)	Supported
BDA→ECV	H3a	0.29	$p < 0.05$ (*)	Supported
BDA→SOV	H3b	0.33	$p < 0.01$ (**)	Supported
IOA→ECV	H4a	0.30	$p < 0.05$ (*)	Supported
IOA→SOV	H4b	0.27	$p < 0.05$ (*)	Supported
BCA→ECV	H5a	0.01	$p > 0.05$ (ns)	Non-Supported
BCA→SOV	H5b	0.21	$p < 0.05$ (*)	Supported
ECV→SMP	H6	0.42	$p < 0.001$ (***)	Supported
SOV→SMP	H7	0.44	$p < 0.001$ (***)	Supported
(ECV→SMP) × EO	H8a	0.16	$p < 0.05$ (*)	Supported
(SOV→SMP) × EO	H8b	0.12	$p < 0.01$ (**)	Supported

Note: $p < 0.05$ (*); $p < 0.01$ (**); $p < 0.001$ (***); $p > 0.05$ (ns).

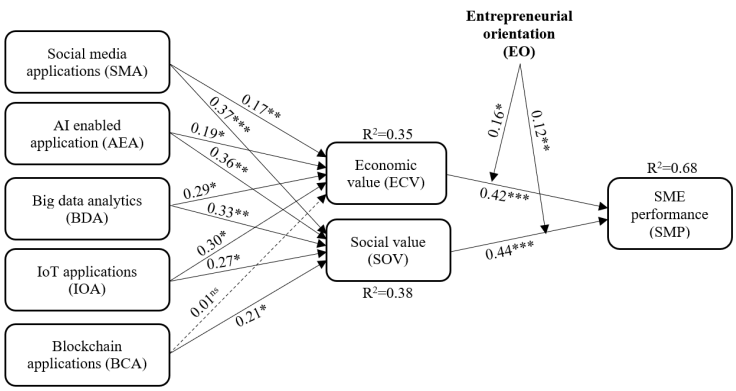


Figure 2. Validated model (SEM). [Note: $p < 0.05$ (*); $p < 0.01$ (**); $p < 0.001$ (***); $p > 0.05$ (ns)].

5.3. Results

The present study has formulated 14 hypotheses and validated them through a statistical process. Out of these 14 hypotheses, two belong to the effects of moderator EO on H6 and on H7. The results demonstrate that SMA significantly and positively impacts both ECV and SOV (H1a and H1b) since the concerned path coefficients are 0.17 and 0.37, and their levels of significance are $p < 0.01$ (**) and $p < 0.001$ (***). Furthermore, AEA significantly and positively impacts both ECV and SOV (H2a and H2b) since the respective path coefficients are 0.19 and 0.36 with respective levels of significance as $p < 0.05$ (*) and $p < 0.01$ (**). The findings also show that BDA impacts ECV and SOV (H3a and H3b) significantly and positively since the concerned path coefficients are 0.29 and 0.33 with respective levels of significance as $p < 0.05$ (*) and $p < 0.01$ (**), and IOA impacts ECV and SOV (H4a and H4b) significantly and positively since the concerned path coefficients are 0.30 and 0.27 with respective levels of significance as $p < 0.05$ (*) and $p < 0.05$ (*). It can be observed that BCA has an insignificant impact on ECV (H5a), as the path coefficient is too low at 0.01 with a non-significance level of $p > 0.05$(ns). However, BCA impacts SOV significantly and positively (H5b) since the concerned path coefficient is 0.21 with a level of significance as $p < 0.05$ (*). This study also presents that ECV and SOV could separately impact SMP (H6 and H7) significantly and positively since the path coefficients are 0.42 and 0.44, respectively, with corresponding levels of significance as $p < 0.001$ (***) and $p < 0.001$ (***). The moderator EO impacts the relationship between H6 and H7 significantly and positively (H8a and H8b) since the concerned path coefficients are 0.16 and 0.12 with respective levels of significance as $p < 0.5$ (*) and $p < 0.01$ (**). Regarding the coefficients of determination (R^2), the results indicate that SMA, AEA, BDA, IOA, and BCA could explain both ECV and SOV as much as 35% ($R^2 = 0.35$) and 38% ($R^2 = 0.38$). The study also revealed that ECV and SOV could simultaneously explain SMP to the extent of 68% ($R^2 = 0.68$), which is the predictive power of the proposed theoretical model.

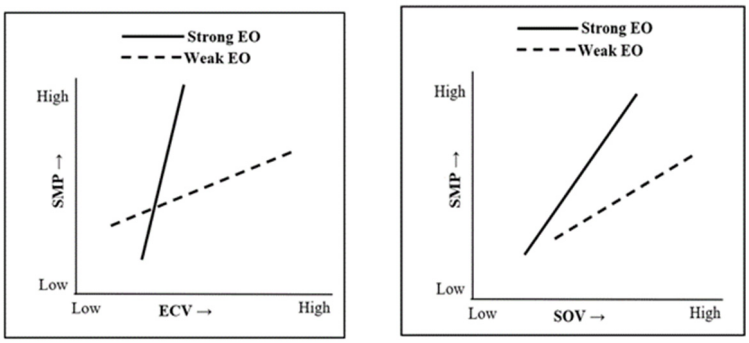
6. Conclusions

The present study has documented how digital transformation, being a strategy of start-up SMEs, has been gaining attention from scholars and practitioners. SMEs are continuously challenged to improve their business abilities and processes. This study has demonstrated that digital transformation in SMEs is able to stimulate new modes of interactions and functions with potential customers, and it has been able to drive SMEs to create new business values. The results show that most digital technologies, such as social media applications, AI, big data analytics, IoT technology, and blockchain applications, can significantly and positively impact the economic and social values of SMEs, which eventually affect SME performance. It is noteworthy that the present study has documented that blockchain technology does not impact ECV (H5a). That is, BCA has an insignificant effect on the economic growth of SMEs, which contradicts the findings from Akter et al. [90]. This is presumably because the results of the present study are based on the analysis of inputs from respondents in India, where the wide application of blockchain technology is not very developed in the SME sector. Digital transformation implementation needs to emphasize how to integrate these emerging technologies in the context of various business functions towards hybrid modes, recombination, integration, as well as in convergence. The present study has put forward that all these digital technologies act as the basic building block for the enterprises towards their future digital transformation journey. This study has dealt with the critical question for the enterprises to establish interconnectivity amongst these emerging technologies for harnessing the eventual benefits. Moreover, the proposed model is expected to put food for reflection not only on the SMEs but also on the other types of enterprises intending to be involved in the digitalization journey.

6.1. Discussion

The present study has demonstrated that SMA, AEA, BDA, IOA, and BCA significantly and positively impact ECV and SOV (except BCA→ECV), which received support

from another study by Akter et al. (2020) [90] that discussed how applications of different digital technologies could transform business styles. The present study has considered the significant moderating effects of EO on the relationships between SMP and its two predictors. The moderating impacts of EO on these two linkages (H6 and H7) have been found significant in terms of MGA, which has also been supported by another study by Diabate et al. [91]. However, the moderating effects of Strong EO and Weak EO on the two linkages covered by H6 and H7 are discussed here with graphical presentations, which are seen in Figure 3.



(**a**) (**b**)

Figure 3. Effects of EO on H6 (**a**) and H7 (**b**).

In both graphs, the continuous lines and dotted lines represent the effects of Strong EO and the effects of Weak EO, respectively. As ECV (for H6) and SOV (for H7) increase, Strong EO causes the rates of increase in SMP in both cases to be greater than compared to the effects of Weak EO. This is because, in both graphs, it appears that the gradients of the dotted lines are less than the gradients of the continuous lines.

6.2. Contributions and Implications

6.2.1. Theoretical Contributions

The present study has provided several theoretical contributions to the extant literature. For example, the findings of the present study demonstrate that, although SMEs could derive individual benefits from the five kinds of technologies, they could derive more economical and social values by effectively harnessing their close interconnectivity to accelerate business growth as well as productivity. The present study then demonstrates that barring a single instance (H5a), these five technologies have accelerated the development of transformative business models. Using these digital applications, SMEs can automate processes, automatically match both demand and supply, and make accurate real-time decisions.

The present study has also extended the concept of RBV theory by arguing that the performance of SMEs is considered a function of their resource mix. It has also been argued that when SMEs have resources that are heterogeneous, specific, and difficult to replicate, they can achieve a better competitive advantage. By using digital applications, enterprises can also create more value for potential customers compared to their competitors. The difference in the performance of enterprises emerges from the variation of their resource portfolios. The five technological resources that have been discussed are valuable, rare, inimitable, and non-substitutable resources, as they seem to provide start-up SMEs with the capacity to implement strategies to enhance their effectiveness and efficiency. The simultaneous use of these five applications can create synergistic benefits, which no other combination that SMEs used could match. Therefore, SMEs achieve superior performance by creating values, which corroborates the extended concept of RBV theory.

These technologies have digitally transformed SMEs, and they are considered important dynamic resources since using these technologies simultaneously could help SMEs to sense the opportunities and seize them for the appropriate benefits. In this way, by extend-

ing the concept of DCV theory, it has been possible to consider the technological abilities of start-up SMEs as their dynamic ability. By using these applications together, start-up SMEs could appropriately react and respond to the high velocity, volatile market environment.

Akter et al. [90] demonstrated the contributions of digital transformation through the lens of some emerging technologies such as AI, IoT, and blockchain. This study also investigated the value propositions of these increasingly converging technologies and applications. That impact has been extended in the present study to investigate how SMEs integrate, converge, recombine, and hybridize five digital technologies to ensure diverse and wide-reaching consequences in various functionalities of their operations. This is claimed to have added value to extant literature.

Another study by Diabate et al. (2019) investigated the effects of EO and its ability on SMEs for sustainable growth in a middle-income economy covering the West African region [91]. By using data extracted from 320 Ivorian SMEs, the study found a close correlation between the effects of EO on the business growth of SMEs. The present study has extended this concept to investigate how applying different digital technologies with the moderating impacts of EO could help SMEs in developing countries improve their resilience and digital entrepreneurship to sustain their growth continuum. This has added value to the body of extant literature.

6.2.2. Implication of Practice

The present study has provided several practical implications. The findings present some fruitful guidelines to managers and leaders of SMEs who intend to digitally transform their companies by using digital technologies such as AI, blockchain, BDA, IoT, and so on. Before investing in developing these technological capabilities, managers and leaders of start-up SMEs need to identify and evaluate if the SMEs can appropriately sense the dynamic changes in the context of changing internal as well as external environments. This may help start-up SMEs to avail themselves of the opportunities and mitigate the risks. Thus, for the digitalization of start-up SMEs, it is essential to know the ability of the employees of the SMEs in the context of their digital maturity level. The start-up SMEs should take a technologically savvy partner who could guide and advise on taking the best strategy through the partner's services and training.

The managers and leaders of SMEs should ascertain if their enterprise possesses the capability to seize the sensed opportunities. They also need to assess if their enterprise has the ability to reconfigure their existing and acquired intangible and tangible assets to successfully create economic and social values to improve firm performance. The managers and leaders of SMEs should also have the patience and foresight to aptly decide when, as well as how, to develop their capabilities and how to explore and exploit their abilities to extract the best potential by simultaneously using digital technologies.

Managers of SMEs should arrange to properly train their employees to appropriately use these technologies and to be motivated to adhere to the environmental obligations that SMEs must obey. This will help SMEs to create economic and social values, which will ensure better performance.

Davenport (2018) opined that people do not fully trust AI-related decision-making processes for services such as medical diagnoses, financial planning, and hiring [92]. This delivers a message that people believe that autonomous systems could replace the people who are employed for such services, but this has not happened yet. In such a scenario, it is argued that it is better to conceptualize that AI will add to human intelligence and not supplant human intelligence, as reported by Carpenter (2015) [93]. In this context, the leadership of SMEs, by improving their entrepreneurial orientation, should disclose that they use a hybrid system (people and machines) and the roles that humans and machines play in the enterprises. This is because the majority of customers have a negative perception of bots and other virtual assistance devices [90].

The application of blockchain and other technologies can create value for SMEs. Since trusts, assets, ownership, contracts, and identity are all stored in the domain of blockchain,

managers of SMEs should try to manage how to successfully capture and create value from each of these components.

The present study has shown that the use of big data analytics eventually helps SMEs to perform better. From this perspective, SMEs need to address the challenges in both technological as well as managerial contexts for successfully extracting value from the huge volume of data [94]. Thus, managers and leaders of SMEs should focus on how digital transformation can successfully integrate all these modern technologies to achieve a better overall performance of the SMEs.

6.3. Limitations and Future Scope

Although the present study has provided some practical and theoretical implications, it is not free from all limitations. The findings depend on such data, which are cross-sectional, which creates defects of causality in the relationship between the constructs and invites endogeneity defects. To eliminate these defects, future researchers may conduct longitudinal studies.

The present study has arrived at the findings based on analyzing data that were obtained from the inputs of respondents in India. The results, therefore, invite external validity issues. It is suggested that future researchers should collect data from respondents dispersed around the world for more generalizable results. The present study arrived at a finding by analyzing the inputs of 319 respondents. The obtained results can hardly be generalized to a large population. Future researchers should analyze the data of more respondents so that the results thus obtained could safely be generalized.

The present study has applied DCV theory, but this theory suffers from the defect of context insensitivity [95]. DCV theory is not capable of identifying the specific condition under which the ability of an SME will be most valuable [96]. In this context, it is suggested that future studies can explore the optimum conditions in which the simultaneous effect of all these modern technologies could create potential superior values for SMEs and ensure better performance. The predictive power of the proposed theoretical model is 68%. It is suggested that future researchers may consider including other boundary conditions and constructs to examine if they could strengthen the predictive power of the model.

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Appendix A

Table A1. A Summary of Questionnaire.

Items	Source (s)	Statements	Response [SD][D][N][A][SA]
SMA1	[36,37]	Social media applications are very popular among younger people.	[1][2][3][4][5]
SMA2	[38,40,42]	I believe different social media applications provide business value to our enterprise.	[1][2][3][4][5]

Table A1. *Cont.*

Items	Source (s)	Statements	Response [SD][D][N][A][SA]
SMA3	[38,41]	I think SMEs are dependent on social media for fulfilling their marketing requirements.	[1][2][3][4][5]
SMA4	[39,42]	I believe that social media helps to develop business activities for the SMEs.	[1][2][3][4][5]
AEA1	[43,44]	SMEs apply AI technologies to help them remodel various business activities.	[1][2][3][4][5]
AEA2	[45,49]	I believe applications of AI can help in supply chain activities of SMEs.	[1][2][3][4][5]
AEA3	[11,46]	Applications of AI can reduce the operational cost of SMEs.	[1][2][3][4][5]
AEA4	[47,48]	I believe that SMEs can use AI applications to develop their customer interaction process.	[1][2][3][4][5]
BDA1	[50]	I believe that data analytics has gained huge momentum in recent years.	[1][2][3][4][5]
BDA2	[51]	Application of big data analytics helps in real time analysis of customers' data.	[1][2][3][4][5]
BDA3	[52]	I believe that applications of big data analytics help in decision making process.	[1][2][3][4][5]
BDA4	[52]	I think SMEs should adopt big data analytics technology to gain competitive advantage.	[1][2][3][4][5]
IOA1	[53,54]	IoT can facilitate rapid exchange of data in a real time scenario.	[1][2][3][4][5]
IOA2	[55–57]	I believe that applications of IoT can help in improving the upscaling process in the SMEs.	[1][2][3][4][5]
IOA3	[12,33]	Applications of IoT can provide a scalable information system that helps SMEs to exchange information quickly.	[1][2][3][4][5]
IOA4	[58,59]	Applications of IoT help SMEs to sense, seize, and reconfigure external opportunities.	[1][2][3][4][5]
BCA1	[61]	Blockchain is considered a digital ledger which presents the detailed history of various transactions.	[1][2][3][4][5]
BCA2	[62]	I believe blockchain technology can save operational cost of SMEs.	[1][2][3][4][5]
BCA3	[63,64]	I think applications of blockchain is secured for the SMEs.	[1][2][3][4][5]
BCA4	[65–67]	I believe that SMEs should adopt blockchain technology for gaining competitive advantage.	[1][2][3][4][5]
ECV1	[31]	SMEs can gain economic value by profit maximization.	[1][2][3][4][5]
ECV2	[68,72]	Adoption of different technologies can provide economic value to the SMEs.	[1][2][3][4][5]
ECV3	[69,73]	The economic value changes if the price of the good or the service changes.	[1][2][3][4][5]
ECV4	[70,71]	I believe that SME leadership should focus more on adopting new-edge technologies.	[1][2][3][4][5]
ECV5	[72]	I believe product development cost can be significantly reduced if SMEs adopt appropriate technologies.	[1][2][3][4][5]
SOV1	[74]	SMEs could gain social benefits if they have performed their work to benefit society.	[1][2][3][4][5]
SOV2	[75]	I believe that social value emerges from the concept of corporate social responsibility programs.	[1][2][3][4][5]
SOV3	[37]	Improving the social value is an important aspect of SMEs.	[1][2][3][4][5]
SOV4	[74]	Customers may favor those SMEs that spend more to uplift the society.	[1][2][3][4][5]

Table A1. *Cont.*

Items	Source (s)	Statements	Response [SD][D][N][A][SA]
SOV5	[37]	I believe that social values are shared values among the employees of the SMEs.	[1][2][3][4][5]
SMP1	[35,76]	I believe that performance of the SMEs can be improved by appropriately adopting modern technologies.	[1][2][3][4][5]
SMP2	[77]	The social value of the SMEs can impact the overall performance of SMEs.	[1][2][3][4][5]
SMP3	[78,79]	Leadership support can play a crucial role in improving SME performance.	[1][2][3][4][5]

SD = Strongly Disagree; D = Disagree; N = Neither agree nor disagree; A = Agree; SA = Strongly Agree.

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